

Practical Demo: Corporate-Bond Spread Risk and Scenario Analysis

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Learning Objectives

After completing this demo you should be able to decompose corporate-bond risk into Treasury-rate, spread, default, and liquidity components; calculate spread duration and duration times spread (DTS); compare spread carry with widening risk; and evaluate mild and severe portfolio scenarios.

1 Inputs and Assumptions

All values are **illustrative, market-like inputs**, not current quotes. The portfolio is intentionally small enough to audit by hand. Prices are near par; annual carry is approximated by spread times market value; spread duration is treated as constant; Treasury and spread P&L use first-order duration; PDs are physical one-year probabilities; defaults and mark-to-market spread changes are reported separately to avoid double counting.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

bonds = pd.DataFrame({
    "Bond": ["A utility 2029", "BBB bank 2031", "BBB industrial 2034", "BB telecom 2028"],
    "Market value ($mm)": [12.0, 10.0, 8.0, 5.0],
    "Treasury duration": [4.2, 5.6, 7.1, 3.4],
    "Spread duration": [4.0, 5.1, 6.3, 3.0],
    "Spread (bp)": [78, 142, 185, 410],
    "One-year PD": [0.0015, 0.0040, 0.0070, 0.0250],
    "Recovery rate": [0.40, 0.40, 0.35, 0.30],
    "Liquidity reserve (bp/year)": [8, 15, 22, 45],
})

bonds["Treasury DV01 ($/bp)"] = (
    bonds["Market value ($mm)"] * 1_000_000 * bonds["Treasury duration"] * 0.0001
)
bonds["Spread DV01 ($/bp)"] = (
    bonds["Market value ($mm)"] * 1_000_000 * bonds["Spread duration"] * 0.0001
)
bonds["DTS"] = bonds["Spread duration"] * bonds["Spread (bp)"] / 10_000
bonds["Contribution to DTS"] = (
    bonds["Market value ($mm)"] / bonds["Market value ($mm)"].sum() * bonds["DTS"]
)
bonds.round(4)
```

	Bond	Market value (\$mm)	Treasury duration	Spread duration	Spread (bp)	One-year PD
0	A utility 2029	12.0	4.2	4.0	78	0.0015
1	BBB bank 2031	10.0	5.6	5.1	142	0.0040
2	BBB industrial 2034	8.0	7.1	6.3	185	0.0070
3	BB telecom 2028	5.0	3.4	3.0	410	0.0250

2 Treasury and Spread Decomposition

For small changes,

$$\frac{\Delta P}{P} \approx -D_T \Delta y_T - D_s \Delta s.$$

Treasury duration measures exposure to the benchmark curve. Spread duration measures exposure to the bond's compensation over that curve. DTS rewrites spread risk for a proportional spread shock:

$$DTS = D_s s, \quad \frac{\Delta P}{P} \approx -DTS \frac{\Delta s}{s}.$$

```
portfolio_value = bonds["Market value ($mm)"].sum() * 1_000_000
treasury_dv01 = bonds["Treasury DV01 ($/bp)"].sum()
spread_dv01 = bonds["Spread DV01 ($/bp)"].sum()
portfolio_dts = bonds["Contribution to DTS"].sum()

pd.Series({
    "Portfolio value ($)": portfolio_value,
    "Treasury DV01 ($/bp)": treasury_dv01,
    "Spread DV01 ($/bp)": spread_dv01,
    "Portfolio DTS": portfolio_dts,
}).round(4)
```

```
Portfolio value ($)      3.5000e+07
Treasury DV01 ($/bp)    1.8020e+04
Spread DV01 ($/bp)      1.6440e+04
Portfolio DTS            7.5600e-02
dtype: float64
```

Equal market values do not imply equal credit risk. Wider-spread bonds can contribute disproportionately to DTS even when their spread duration is shorter.

3 Carry Versus Widening Breakeven

Approximate annual spread carry is $MV \times s$. The spread widening that consumes one year of spread carry is

$$\Delta s_{BE} \approx \frac{s}{D_s}.$$

This is a mark-to-market breakeven, not a total-return forecast. It ignores rolldown, Treasury carry, default, taxes, and financing.

```
bonds["Annual spread carry ($)"] = (  
    bonds["Market value ($mm)"] * 1_000_000 * bonds["Spread (bp)"] / 10_000  
)  
bonds["Widening breakeven (bp)"] = bonds["Spread (bp)"] / bonds["Spread duration"]  
bonds[["Bond", "Annual spread carry ($)", "Widening breakeven (bp)"]].round(1)
```

	Bond	Annual spread carry (\$)	Widening breakeven (bp)
0	A utility 2029	93600.0	19.5
1	BBB bank 2031	142000.0	27.8
2	BBB industrial 2034	148000.0	29.4
3	BB telecom 2028	205000.0	136.7

4 Expected Loss and Liquidity Reserve

Expected credit loss is

$$EL = PD \times LGD \times EAD, \quad LGD = 1 - \text{Recovery Rate}.$$

Spread is not expected loss. It may also compensate for systematic credit risk, uncertainty, liquidity, taxes, and supply-demand effects.

```
bonds["LGD"] = 1 - bonds["Recovery rate"]  
bonds["Expected loss ($)"] = (  
    bonds["Market value ($mm)"] * 1_000_000 * bonds["One-year PD"] * bonds["LGD"]  
)  
bonds["Liquidity reserve ($)"] = (  
    bonds["Market value ($mm)"] * 1_000_000 * bonds["Liquidity reserve (bp/year)"] / 10_000  
)
```

```

bonds["Carry after EL and liquidity ($)"] = (
    bonds["Annual spread carry ($)"]
    - bonds["Expected loss ($)"]
    - bonds["Liquidity reserve ($)"]
)
bonds[["Bond", "Annual spread carry ($)", "Expected loss ($)",
        "Liquidity reserve ($)", "Carry after EL and liquidity ($)"]].round(0)

```

	Bond	Annual spread carry (\$)	Expected loss (\$)	Liquidity reserve (\$)	Carry after EL
0	A utility 2029	93600.0	10800.0	9600.0	73200.0
1	BBB bank 2031	142000.0	24000.0	15000.0	103000.0
2	BBB industrial 2034	148000.0	36400.0	17600.0	94000.0
3	BB telecom 2028	205000.0	87500.0	22500.0	95000.0

5 Mild and Severe Scenarios

The scenarios distinguish a benchmark-rate shock from issuer spread shocks. Expected loss is shown as a separate one-year planning measure rather than added to scenario P&L.

```

scenarios = {
    "Mild slowdown": {
        "Treasury shock (bp)": -20,
        "Spread shocks (bp)": np.array([15, 30, 40, 80]),
    },
    "Severe recession": {
        "Treasury shock (bp)": -75,
        "Spread shocks (bp)": np.array([55, 110, 160, 350]),
    },
}

rows = []
for name, scenario in scenarios.items():
    rate_pnl = -bonds["Treasury DV01 ($/bp)"] * scenario["Treasury shock (bp)"]
    spread_pnl = -bonds["Spread DV01 ($/bp)"] * scenario["Spread shocks (bp)"]
    rows.append({
        "Scenario": name,
        "Treasury P&L ($)": rate_pnl.sum(),
        "Spread P&L ($)": spread_pnl.sum(),
        "Total mark-to-market P&L ($)": rate_pnl.sum() + spread_pnl.sum(),
    })

```

```
} )
```

```
scenario_table = pd.DataFrame(rows).set_index("Scenario")
scenario_table.round(0)
```

	Treasury P&L (\$)	Spread P&L (\$)	Total mark-to-market P&L (\$)
Scenario			
Mild slowdown	360400.0	-546600.0	-186200.0
Severe recession	1351500.0	-2156400.0	-804900.0

```
plot_data = scenario_table[["Treasury P&L ($)", "Spread P&L ($)"]] / 1_000
ax = plot_data.plot(kind="bar", figsize=(7.4, 4.0), color=["#4C78A8", "#F58518"])
ax.axhline(0, color="black", linewidth=0.8)
ax.set(ylabel="P&L ($ thousands)", title="Scenario P&L decomposition")
ax.tick_params(axis="x", rotation=0)
ax.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()
```

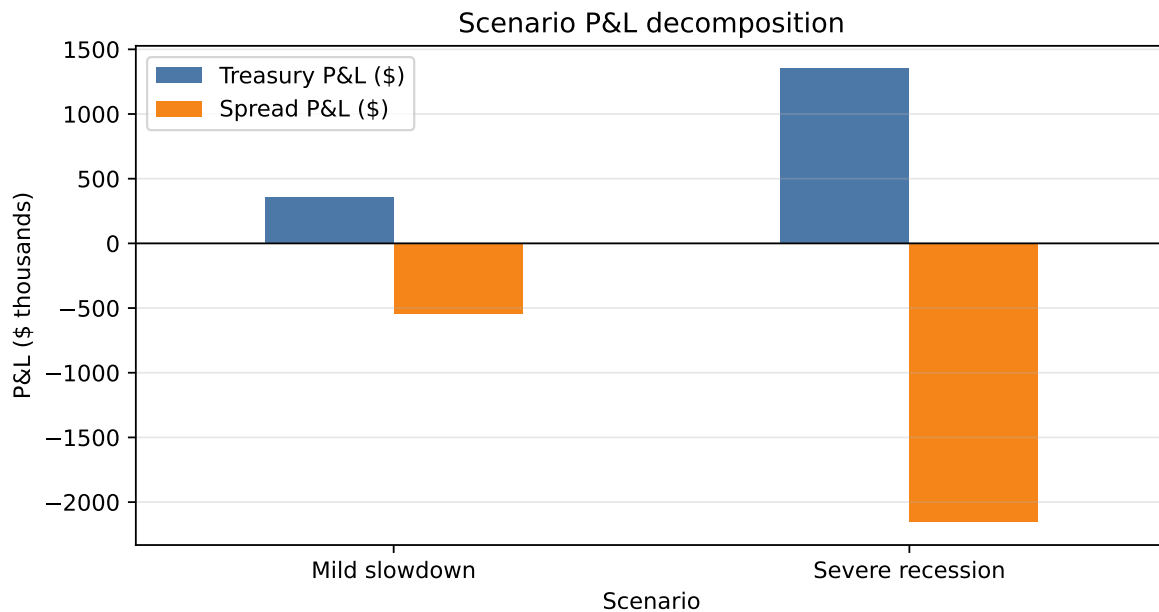


Figure 1: Falling Treasury yields can partly offset, but do not eliminate, spread losses

6 Why the Risks Differ

- **Rate risk** is exposure to the benchmark Treasury curve and can often be hedged with Treasury futures or swaps.
- **Spread risk** is mark-to-market exposure to changing required compensation for credit and liquidity risk.
- **Default risk** is a discrete credit event with uncertain recovery; PD and LGD describe expected loss, not daily spread volatility.
- **Liquidity risk** is the cost or inability to trade near an observed price, often worst when spreads are already widening.

7 Validation Checks

```
assert np.all(bonds["DTS"] > 0)
assert np.all(bonds["Widening breakeven (bp)"] > 0)
assert np.all(bonds["Expected loss ($)"] >= 0)
assert scenario_table.loc["Severe recession", "Spread P&L ($)"] < scenario_table.loc["Mild s
assert abs(portfolio_dts - np.average(bonds["DTS"], weights=bonds["Market value ($mm)"])) <

pd.Series({
    "DTS is positive": bool(np.all(bonds["DTS"] > 0)),
    "Expected loss is nonnegative": bool(np.all(bonds["Expected loss ($)"] >= 0)),
    "Severe spread loss exceeds mild spread loss": True,
    "Portfolio DTS reconciles to weighted bond DTS": True,
})
```

DTS is positive	True
Expected loss is nonnegative	True
Severe spread loss exceeds mild spread loss	True
Portfolio DTS reconciles to weighted bond DTS	True

dtype: bool

8 Financial Interpretation

A wide spread is not automatically attractive. The analyst should ask how much of the spread may be consumed by expected loss and liquidity, how much widening erases carry, and whether falling Treasury yields are likely to offset spread losses in stress. DTS is useful for

proportional spread shocks; DV01 is useful for basis-point shocks; neither replaces explicit default and liquidity analysis.